

- Introduction to data: what is data?

Meaning of statistics in real life

Statistics is the study of collecting, organizing, analyzing, and interpreting data to help us understand information and make decisions.

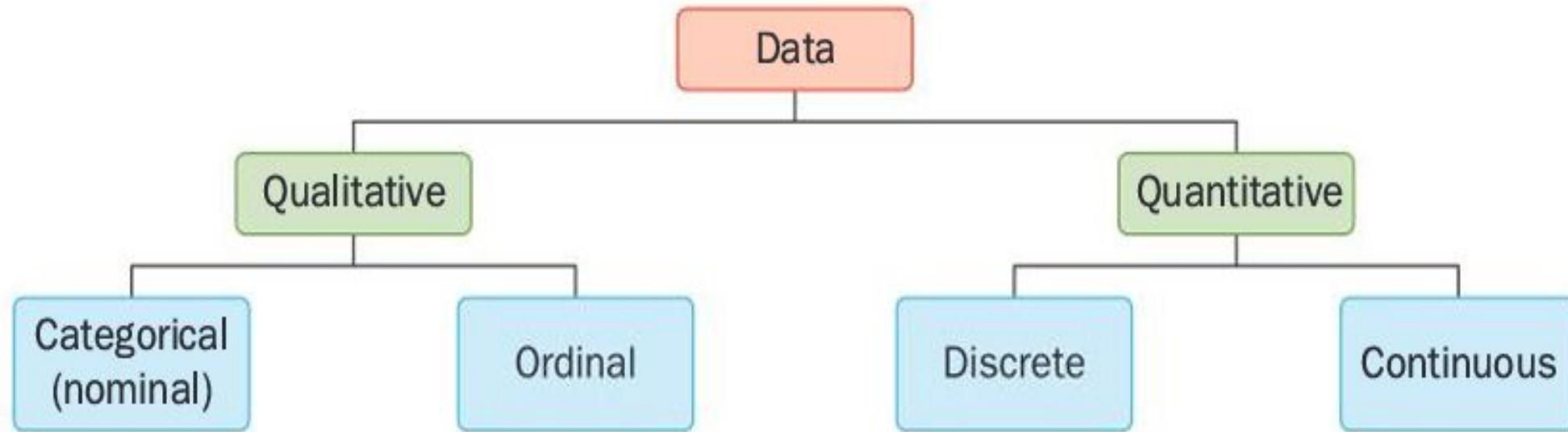
Data:

Data is a collection of information, facts, or numbers gathered through observation, counting, or measurement.



Types of Data

Data can be classified in the following ways.



Quantitative data contain values or numbers (quantities), whereas *qualitative data* represent words or descriptions (qualities).

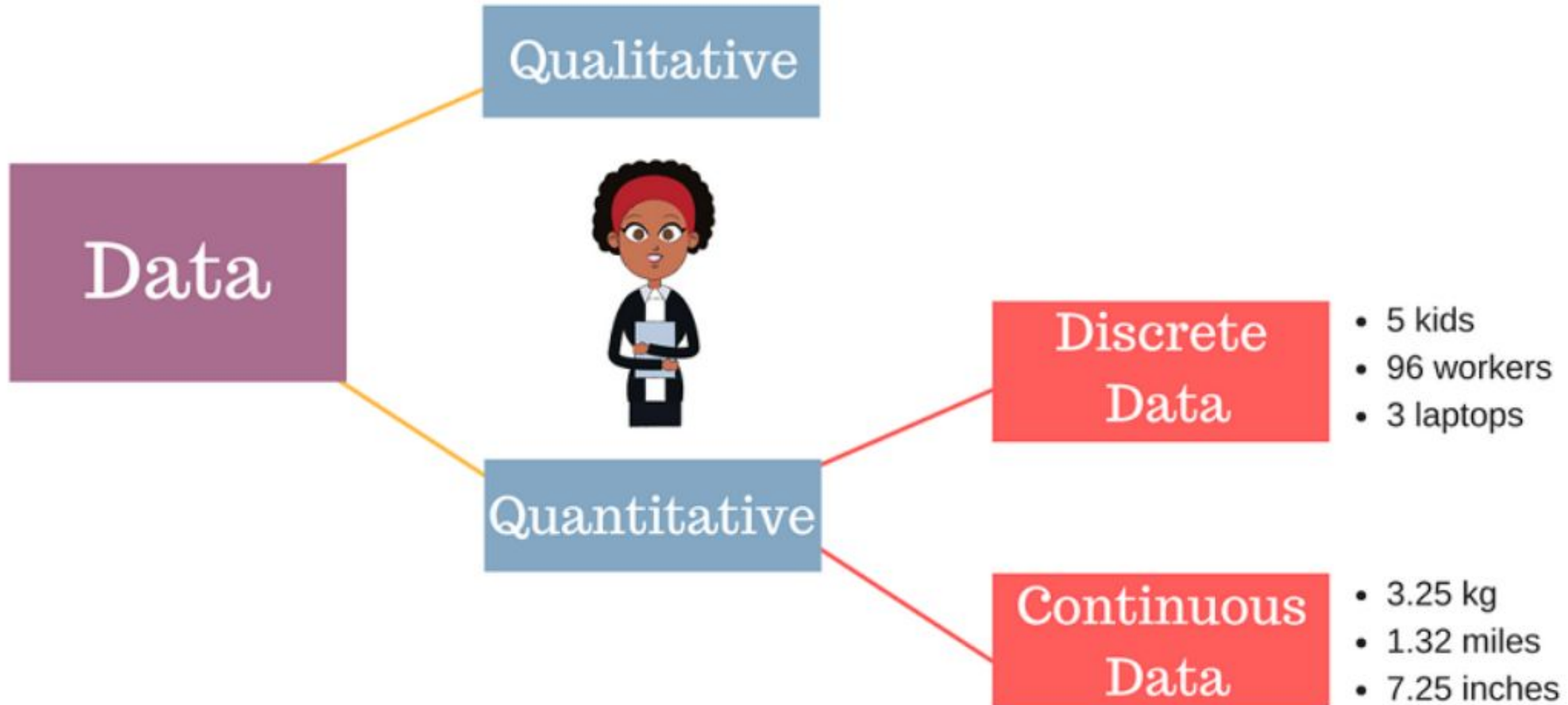


Types of Qualitative Data

What is Qualitative data?	Non numerical information that captures the qualities & characteristics of a phenomenon. Expressed through words, texts, images, or observations.	
Types	Nominal	Ordinal
Definition	Categorical data that show labels without any order.	Ordered non numerical data that describe attributes & categories.
Examples	Hair colors, types of cars, gender, types of pets, religion, marital status.	Education level, dress size, customer satisfaction rate, pain scale.



Types of Data



Definitions:

Qualitative vs quantitative data:

- Qualitative data describes qualities or characteristics.
It is non-numerical data.

Examples: Favourite colour

- Quantitative data is numerical data that can be measured or counted.

Examples: Age of students



Definitions:

Discrete vs continuous data

- Discrete data is countable and has whole number values only.

Examples: Number of students in a class

- Continuous data is measurable and can take any value, including decimals.

Examples: Height of a student



Discrete data key characteristics:

- **You can count the data.** It is usually units counted in whole numbers.
- **The values cannot be divided** into smaller pieces and add additional meaning.
- **You cannot measure the data.** By nature, discrete data cannot be measured at all. For example, you can measure your weight with the help of a scale. So, your weight is not a discrete data.
- **It has a limited number of possible values** e.g. days of the month.
- Discrete data is **graphically displayed by a bar graph.**

- The number of students in a class.
- The number of workers in a company.
- The number of parts damaged during transportation.
- Shoe sizes.
- Number of languages an individual speaks.
- The number of home runs in a baseball game.
- The number of test questions you answered correctly.
- Instruments in a shelf.
- The number of siblings a randomly selected individual has.



Continuous data key characteristics:

- In general, **continuous variables are not counted**.
- **The values can be subdivided** into smaller and smaller pieces and they have additional meaning.
- The continuous **data is measurable**.
- It has an **infinite number of possible values** within an interval.
- Continuous data is **graphically displayed by histograms**.



- The amount of time required to complete a project.
- The height of children.
- The amount of time it takes to sell shoes.
- The amount of rain, in inches, that falls in a storm.
- The square footage of a two-bedroom house.
- The weight of a truck.
- The speed of cars.
- Time to wake up.

Data Collection Methods

Primary



Survey/
Questionnaire



Interview



Observation



Experiment

Secondary



Literature
Review



Government
Database



Commercial
Database



Web
Scraping



Sources of data: Sources of data tell us where data comes from.

Data can be collected in two main ways:

- **Primary data:** Primary data is data that is collected first-hand by the researcher for a specific purpose.

Examples: Conducting a class survey

- **Secondary data:** Secondary data is data that is already collected by someone else and is used again.

Examples: Data from the internet



Methods of data collection:

Methods of data collection are the ways we gather data to answer questions or solve problems.

- **Surveys:** A survey is a method of collecting data by asking questions to a group of people.

Example: Asking students about their favourite sport

- **Questionnaires:** A questionnaire is a set of written questions used in a survey.

Example: A form asking: How many hours do you study daily?

- **Observation:** Observation is a method of collecting data by watching and recording events or behaviour.

Example: Observing how many students bring lunch from home

